WEEK 6- TASK B: PROJECT PROPOSAL

Section 1: Track Chosen

Track B — Fluid Dynamics & Navier–Stokes with Physics-Informed Neural Networks

Section 2: Project Description

1. **Problem Statement:** This project aims to solve the steady two-dimensional incompressible Navier–Stokes equations for the classical lid-driven cavity flow using Physics-Informed Neural Networks (PINNs). Unlike conventional CFD methods that discretize the governing equations on a mesh, the PINN will directly learn the velocity and pressure fields by minimizing the residuals of the Navier–Stokes equations and enforcing the prescribed boundary conditions.
2. **Implementation:** The project will be implemented using PyTorch in Python Jupyter Notebook. The neural network will predict the velocity components (u,vu, vu,v) as functions of the spatial coordinates (x,y)(x, y)(x,y). Automatic differentiation will be used to compute the required first- and second-order derivatives for constructing the continuity and momentum residuals. The total training loss will combine the Navier–Stokes residuals with boundary-condition losses. To improve convergence and address optimization challenges associated with PINNs, techniques such as Fourier Feature Embedding and Gradient-Norm Balancing will be incorporated.OpenFOAM software, along with its visualizer PARAVIEW, has to be installed. It contains several conventional CFD solvers for canonical problems. The resulting PINN solution of the velocity fields will be benchmarked against reference solutions generated using conventional OpenFOAM CFD solvers or canonical problems, which is in this case, the 2D-lid driven cavity problem. Next, the pressure recovery will be validated by comparing the neural network results again with those from the OpenFOAM solver. Also, sparse sensor data assimilation will be implemented, in which some velocity values from the OpenFOAM solver will be fed into the neural network, enabling it to learn from data. There is also future scope to implement a custom PiNN-based solver within OpenFOAM itself, rather than using Python Notebook.
3. **Expected Result:** The trained PINN is expected to accurately reconstruct the steady velocity and pressure fields while satisfying the incompressibility constraint throughout the computational domain. Furthermore, the usage of sparse sensor data should improve the performance of the PiNN, bringing down the L^2 value.

Section 3: Reading Plan

To obtain a holistic understanding of PiNNs for CFD, I will read the following papers to review recent developments, practical challenges, and applications of PiNNs in computational fluid dynamics.

[1] S. Cai, Z. Mao, Z. Wang, M. Yin, and G. E. Karniadakis, *"Physics-informed neural networks (PINNs) for fluid mechanics: A review,"* arXiv preprint arXiv:2105.09506, 2021. Available: https://arxiv.org/abs/2105.09506

[2] X. Jin, S. Cai, H. Li, and G. E. Karniadakis, *"NSFnets (Navier–Stokes flow nets): Physics-informed neural networks for the incompressible Navier–Stokes equations,"* **Journal of Computational Physics**, vol. 426, p. 109951, Feb. 2021. doi: 10.1016/j.jcp.2020.109951. Available: https://arxiv.org/abs/2003.06496

[3] M. Raissi, A. Yazdani, and G. E. Karniadakis, *"Hidden Fluid Mechanics: A Navier–Stokes Informed Deep Learning Framework for Assimilating Flow Visualization Data,"* arXiv preprint arXiv:1808.04327, 2018. Available: https://arxiv.org/abs/1808.04327